

Normal Distribution

The Normal Distribution is an important continuous probability distribution used to model various natural and real-life phenomena such as human heights, weights, and measurement errors. It is characterized by a symmetric, bell-shaped curve and is completely determined by two parameters:

- Mean = (μ) ,
- Standard deviation = (σ) .

Probability Density Function (PDF)

The probability density function of a normal distribution with mean μ and standard deviation σ is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty, \sigma > 0.$$

Properties of Normal Distribution

- The curve is symmetric about the mean μ .
- In a normal distribution, Mean = Median = Mode.
- The total area under the curve is 1.
- The Empirical Rule (68–95–99.7 Rule) holds:
 - Approximately 68% of data lies within 1σ of μ .
 - Approximately 95% of data lies within 2σ of μ .
 - Approximately 99.7% of data lies within 3σ of μ .

Standard Normal Variate

A Standard Normal Distribution is a special case of the normal distribution where:

$$\mu = 0 \quad \text{and} \quad \sigma = 1.$$

If X is a normal random variable with mean μ and standard deviation σ , the standard normal variate Z is defined by:

$$Z = \frac{X - \mu}{\sigma}.$$

This transformation converts any normal distribution into the standard normal distribution, allowing probability values to be obtained from standard normal tables.

Examples

1. X is normally distributed with mean 12 and standard deviation 4. Find:
 - (a) $P(X > 20)$
 - (b) $P(0 < X < 12)$
2. The mean yield per acre is 662 kg with standard deviation 32 kg. In a batch of 1000 plots, how many plots would you expect to have yields:
 - (a) More than 700 kg
 - (b) Less than 650 kg
 - (c) Between 500 kg and 700 kg
3. A machine produces rods whose lengths are normally distributed with mean 50 cm and standard deviation 3 cm. Find the probability that a rod is:

- (a) Longer than 56 cm
 - (b) Between 45 cm and 50 cm
4. Heights of students are normally distributed with mean 160 cm and standard deviation 8 cm. What proportion of students are taller than 175 cm?
 5. The lifetime of a bulb follows a normal distribution with mean 1200 hours and standard deviation 150 hours. Find the probability that a bulb lasts between 1000 and 1400 hours.
 6. The test scores in an exam follow a normal distribution with mean 60 and standard deviation 10. What percentage of students scored:
 - (a) Above 75
 - (b) Below 45
 7. Daily electricity consumption in a city follows $N(250, 36)$. Find $P(200 < X < 300)$.
 8. A worker's daily earnings are normally distributed with mean \$550 and standard deviation \$40. Find the probability that on a particular day the worker earns more than \$600.
 9. The weight of apples is normally distributed with mean 180 g and standard deviation 20 g. How many apples out of 500 weigh less than 150 g?
 10. The pressure in a boiler is normally distributed with mean 300 kPa and standard deviation 25 kPa. Find the probability that the pressure exceeds 350 kPa.

Exponential Distribution

The Exponential Distribution is a continuous probability distribution commonly used to model waiting times between independent events occurring at a constant average rate. It is widely applied in reliability engineering, survival analysis, and queuing theory.

Probability Density Function (PDF)

If X is an exponentially distributed random variable with parameter $\theta > 0$, then its probability density function (PDF) is:

$$f(x) = \theta e^{-\theta x}, \quad x \geq 0.$$

Mean and Variance

For the exponential distribution with parameter θ :

$$E[X] = \frac{1}{\theta},$$

$$Var(X) = \frac{1}{\theta^2}.$$

Examples

1. If $X \sim \text{Exponential}(\theta = 2)$, find the mean and variance.
2. The waiting time between phone calls follows an exponential distribution with parameter $\theta = 0.25$. Find $P(X > 5)$.
3. A device has an exponential lifetime with mean 12 hours. Find the value of θ and the variance.
4. If the failure time of a machine follows $\text{Exponential}(\theta = 0.1)$, find $P(10 < X < 20)$.
5. The lifetime of bulbs is exponentially distributed with variance 49. Find θ , the mean, and the PDF of the distribution.

Gamma Distribution (One-Parameter Form)

The Gamma Distribution in one-parameter form is obtained when the shape parameter $\lambda > 0$ is known and the scale parameter is equal to 1. This form is also known as the *standard gamma distribution*.

Probability Density Function (PDF)

If X follows a Gamma distribution with shape parameter λ , then the probability density function is:

$$f(x) = \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)}, \quad x > 0.$$

Here, $\Gamma(\lambda)$ is the Gamma function defined by:

$$\Gamma(\lambda) = \int_0^{\infty} t^{\lambda-1} e^{-t} dt.$$

Mean and Variance

For the one-parameter gamma distribution:

$$\begin{aligned} E[X] &= \lambda, \\ \text{Var}(X) &= \lambda. \end{aligned}$$

Thus, in this distribution, **mean equals variance**.